Can we learn more when we cannot collect more data?

Counterfactual Residual Data Augmentation for Regression

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Authors' demo run on 3 laptops · Table 1 not yet replicated

Anik · Alvi · Atul

Small, noisy tables are hard to learn from

01 Few samples

02 Costly to collect more

03 Noisy observations

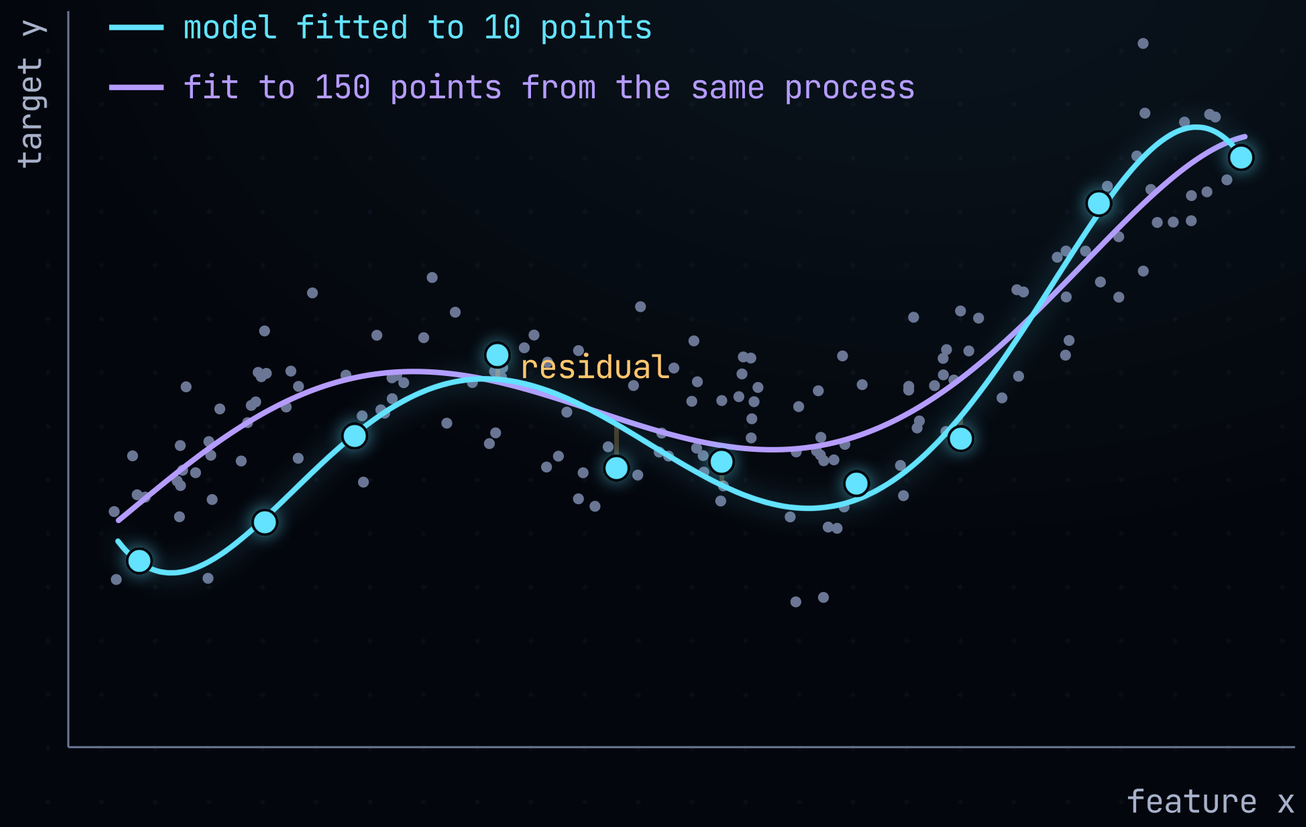
REGRESSION

Predict a *number* from a row of features — e.g. a building's heating load from its geometry.

RESIDUAL

What the model got wrong for that row: observed − predicted.

$$z_i = y_i - \hat{g}(x_i)$$



Illustrative example — synthetic points, real least-squares fits

Images have label-safe transformations. Tables rarely do.

Images: label-preserving for suitable transforms



LABEL “7” ✓ small rotations keep it a 7

…but rotate a 6 by 180° and it reads as a 9

Tabular regression: no known label-safe transform

FLOOR AREA	BEDS	GARAGE	LOCATION	PRICE (TARGET)
92 m²	3	basic	zone B	£285,000
110 m²	3	basic	zone B	£285,000

NAIVE COPY floor area +20%, price copied over

✗ Unjustified label: nothing tells us what this house would sell for

Illustrative example — invented values

INTERPOLATE ROWS

SMOTE-style, C-Mixup (Chawla et al., 2002; Yao et al., 2022)

GENERATE ROWS

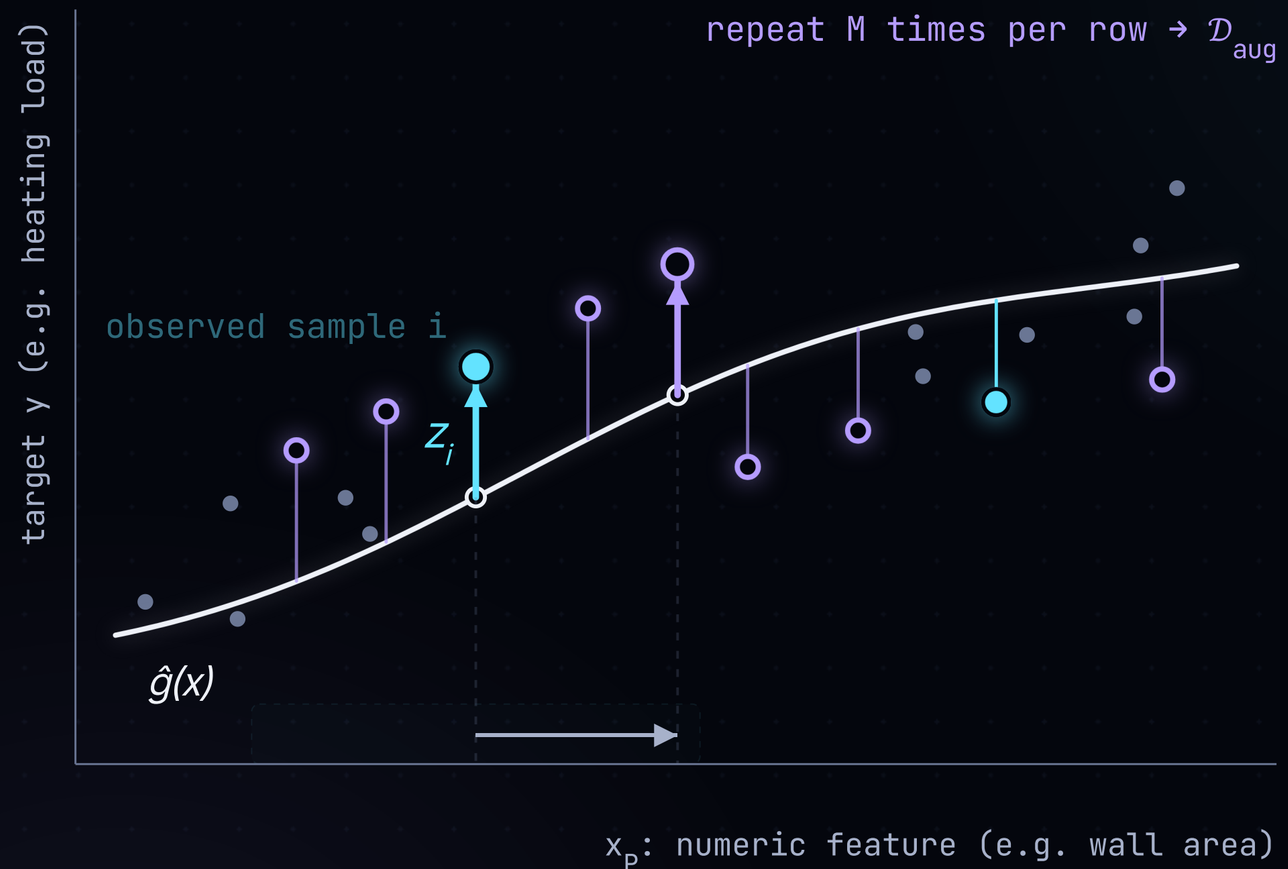
CTGAN, TabDDPM — target treated as just another column (Xu et al., 2019; Kotelnikov et al., 2023)

AUTHORS’ CRITIQUE

can help in places, but don’t keep each row’s *own* residual noise; can raise MSE in their benchmarks

Row: illustrative, invented values · Approaches as characterised by Mohebbi et al. (2026), §2 & Table 2

Move a safe numeric feature. Keep the observed noise.



Illustrative sketch — invented points, not paper data. One feature shown; CRDA may scale several screened numeric features.

Paper's analogy (§1): change a house's garage finish, keep its "bidding-war" residual.

1. **Fit a baseline predictor**
   $\hat{g} \leftarrow \text{MLP or XGBoost on } \mathcal{D}_{\text{train}}$
2. **Compute the original residual**
   $z_i = y_i - \hat{g}(x_i)$
3. **Scale selected numeric features**
   $x'_{i,P} = x_{i,P}\,(1+\delta), \quad \delta \sim \mathcal{U}[-p, p]$
   $x_{i,R}$ held fixed · repo default p = 0.1 · shown: δ = +0.09
4. **Reuse the same residual**
   $y'_i = \hat{g}(x'_i) + z_i$
5. **Repeat M times per row**
   $\mathcal{D}_{\text{aug}} = \{(x'_i, y'_i)\}$

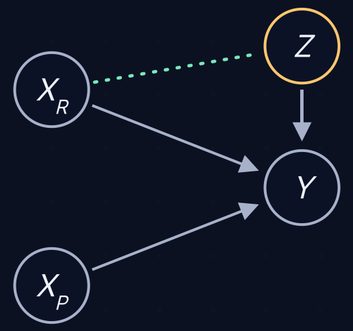
04 — WHAT MUST BE TRUE

It only works if the noise ignores the features we move

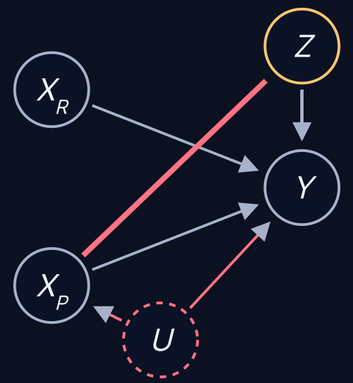
$$P(Z \mid X_P, X_R) = P(Z \mid X_R)$$

Given the fixed features, the residual carries no information about the perturbed ones.

(A) Theory · Assumption 3.1 — assumed, not guaranteed by the algorithm



✓ holds X_P moves Y, not Z. Z may depend on X_R.



✗ breaks hidden U drives X_P and Y; Z absorbs U.

→ causal effect ⋯ allowed dependence ━ violates 3.1 after paper Fig. 1

(B) HEURISTIC SCREENS — EVIDENCE, NOT PROOF

F1 ~~F2~~ F3 F4 ~~F5~~ F6 ~~F7~~ F8 illustrative features

PC CAUSAL-GRAPH CHECK
drop features with an edge to $\hat{z}$

PEARSON SCREEN
drop features correlated with $\hat{z}$ (repo default $|r| \geq 0.1$)

survivors → X_P

(C) PREDICTIVE GATE · WILCOXON SIGNED-RANK
10-fold CV errors, augmented vs. unaugmented
p < 0.05 → use CRDA otherwise → keep baseline
Evidence on validation folds — not a guarantee on new data

Also fails if ĝ is inaccurate: its errors leak into ẑ and stay tied to X_P (§3.3).

Screens are heuristics: PC can miss hidden confounders; Pearson misses non-linear links (§6).

Mohebbi et al. (2026), Assump. 3.1, Fig. 1 adapted, §3.3, §6 · Spirtes et al. (2000) · Wilcoxon (1945)

A focused, checkable reproduction — not a re-invention

✓ **Install & audit**
authors' repo, Python 3.11, pinned `requirements.txt`

✓ **Run the demo**
`demo/demonstration.ipynb` — Energy Efficiency

3 **Initial scope** PROPOSED
Energy Efficiency + Concrete Strength · MLP & XGBoost

4 **Controlled comparison**
unaugmented vs CRDA, identical splits; 3 seeds first pass (paper: 15)

5 **Compare with published**
our ΔMSE next to Table 1, matching their protocol

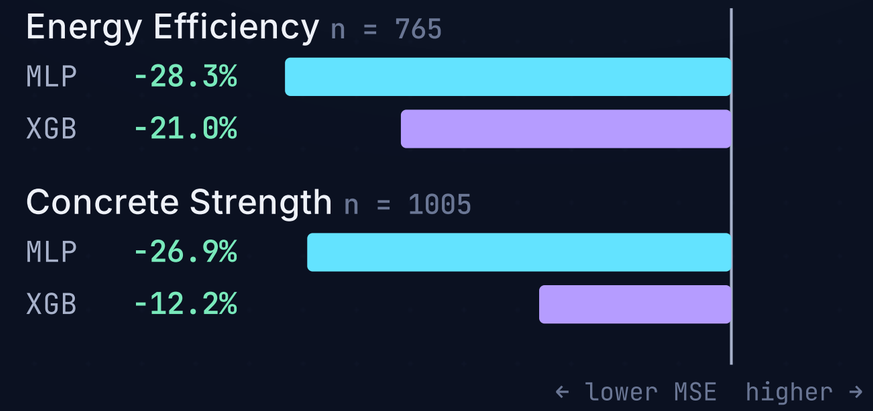
6 **Document**
differences, obstacles, versions, runtimes

STEPS 1–2 DONE ON 3 LAPTOPS · STEPS 3–6 PROPOSED

Original authors' reported results — **not our group's reproduced results**

−22.9% MLP **−6.4%** XGBoost

average test-MSE change across 9 benchmark datasets



Paper Table 1 · mean of 15 seeds · largest subset size

CODE-READING QUESTIONS — LINES FOUND IN THE AUTHORS' REPO; WHAT THEY IMPLY IS NOT YET CHECKED

? Table 1 configs set `ignore_filter: true`, so ΔMSE is measured with augmentation forced on — the Wilcoxon decision is only logged

? Code scores the augmented model as the average of 10 CV-fold models; Alg. 1 describes one retrained model

Reported values: Mohebbi et al. (2026), abstract & Table 1 · Code questions: repo experiments/*/config.json, src/experiment.py

06 — OUR FIRST CHECK

The authors' demo runs on all three of our laptops

One seed · one dataset · CPU only — not Table 1

LAPTOP	RUN TIME	MLP · TEST MSE BEFORE → AFTER	XGBOOST · TEST MSE BEFORE → AFTER
Authors' own notebook output stored in their repo's demo/	—	0.00132 → 0.00064 −51.7% p = 0.00098	0.00130 → 0.00093 −28.5% p = 0.042
Anik Apple M2 (8 cores) · 8 GB GPU: Apple M2 integrated — not used macOS 14.6.1 · Python 3.11.5	41 s	0.00132 → 0.00064 −51.7% p = 0.00098 = authors' output	0.00130 → 0.00093 −28.5% p = 0.042 = authors' output
Alvi AMD Ryzen 7 5800H (8 cores) · 64 GB GPU: NVIDIA RTX 3070 Laptop — not used Windows 11 (build 26340) · Python 3.11.15	3 min 04 s	0.00132 → 0.00064 −51.7% p = 0.00098 = authors' output	0.00127 → 0.00095 −24.8% p = 0.032 differs · baseline 2.7% lower
Atul Apple M5 Pro · 24 GB GPU: Apple M5 Pro integrated (16-core) — not used macOS 27.0.1 · Python 3.11.17	34 s	0.00132 → 0.00064 −51.7% p = 0.00098 = authors' output	0.00130 → 0.00093 −28.5% p = 0.042 = authors' output

PINNED BY THE AUTHORS, SAME ON ALL THREE

torch 2.7.0 · scikit-learn 1.7.0rc1 · xgboost 3.0.1 · numpy 2.2.6

DEVIATIONS LOGGED

! Pinned scipy 1.15.3 won't load on macOS 27 → used 1.16.0 on Atul's Mac; numbers unchanged

! Windows XGBoost: unaugmented baseline already differs (0.00127 vs 0.00130); cause unknown

Clear ownership, balanced load, every result cross-checked

Proposed roles — confirm with team

A

PROPOSED

Anik

Environment & reproducibility

- Python 3.11 env from the authors' pinned requirements
- Environment notes for every platform we use
- Seed, config and version logging

CROSS-CHECK

Re-runs one of Alvi's configs independently

SPEAKS

slides 1, 2, 3, 7 + own card

B

PROPOSED

Alvi

Experiment execution

- Baseline vs CRDA runs (MLP, XGBoost)
- Controlled splits and seeds
- Saves every config.json / results.csv

CROSS-CHECK

Recomputes Atul's comparison table from raw results

SPEAKS

slides 4, 5, 7 + own card

C

PROPOSED

Atul

Analysis, documentation & delivery

- Our ΔMSE next to the paper's Table 1
- Obstacle and deviation log
- Slides, timing and rehearsal

CROSS-CHECK

Rebuilds Anik's environment from the written steps

SPEAKS

slides 6, 7, 9, 10 + own card



Shared by everyone: run at least one config end-to-end · review the final slides · rehearse to time

08 — TARGETS BEFORE WEEK 9

What we will bring to the next update

Targets 2–4 proposed — weeks and owners to confirm

Week 5 · today — Week 9 · progress update

1 — ✓ Done 22 Oct

Environment reproduced

DONE WHEN

Authors' demo runs end-to-end on our laptops

EVIDENCE

Exact versions + CPU/OS logged; 3 laptops (docs/demo-reports/)

Anik

2 — by Week 7

Baselines logged

DONE WHEN

Unaugmented MLP + XGBoost on dataset 1, fixed seeds

EVIDENCE

config.json + results.csv per run

Alvi

3 — by Week 8

First CRDA comparison

DONE WHEN

Our ΔMSE next to the paper's Table 1 row

EVIDENCE

Side-by-side table, Wilcoxon p-values

Alvi Atul

4 — Ongoing

Obstacle log

DONE WHEN

Every deviation recorded with its fix or open question

EVIDENCE

Shared log, discussed at Week 9

Atul

OUR RESEARCH QUESTION

Do CRDA's reported gains hold when we re-run the authors' own pipeline — and where do they not?

Weeks and owners proposed, pending team agreement · target 1 done (22 Oct); targets 2–4 planned, not started

REFERENCES

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All URLs checked 19 October 2026